Adaptive Multi-Phase Negotiation Strategy: An Analysis of the HiHan Agent Architecture

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Abstract

This report provides a comprehensive architectural analysis of the *Han* agent, a sophisticated automated negotiation agent designed for Stacked Alternating Offers (SAO) protocols. The agent employs a robust, time-dependent negotiation strategy that balances aggressive utility maximization with pragmatic concession behaviors to secure profitable agreements. Core features include a Boulware-like concession tactic, strict reservation value enforcement, an AC_next acceptance mechanism, and frequency-based opponent modeling. Furthermore, the agent differentiates itself through a multi-modal, psychologically-driven communication strategy divided into three distinct phases. This document details the mathematical models, algorithmic decision-making processes, and communication paradigms integrated into the Han agent's framework.

1 Introduction

In the domain of multi-agent systems, automated negotiation serves as a critical mechanism for conflict resolution and resource allocation. The Stacked Alternating Offers (SAO) protocol is a standard framework where agents exchange proposals sequentially until an agreement is reached or a deadline expires. Operating within such constraints requires agents to dynamically adapt to relative time (t), estimate the opponent's flexibility, and strategically concede without compromising their baseline utility.

The *Han* agent (implemented as a subclass of `SAONegotiator` within the NegMAS framework) is engineered to address these challenges through a comprehensive, time-dependent strategy. Its design philosophy is rooted in protecting the principal's utility during the early stages of negotiation while ensuring that viable agreements are not lost to time constraints in the final stages. To achieve this, the agent combines a strict mathematical bidding model with a natural-language communication module designed to persuade human or hybrid counterparts.

2 Core Bidding Strategy

The Han agent's bidding mechanism is driven by a non-linear concession model designed to maximize the extracted utility while avoiding negotiation breakdowns.

2.1 Time-Dependent Boulware Concession

At the heart of the agent's bidding strategy is a Boulware-like concession tactic. A Boulware strategy is characterized by extreme stubbornness in the early and middle stages of negotiation, followed by rapid concessions as the deadline approaches.

The agent calculates its target utility (aspiration level) dynamically based on the normalized relative time $t \in [0, 1]$. The aspiration function $A(t)$ is defined as:

$$A(t) = RV + (1.0 - RV) \times (1.0 - t^e) \tag{1}$$

where RV represents the absolute reservation value (the walk-away point) and e is the Boulware exponent. In the Han implementation, the default exponent is strictly set to $e = 4.0$. At $t = 0$, the aspiration is 1.0 (maximum utility). As t approaches 1.0, the aspiration asymptotically drops to the reservation value. This slow initial concession curve ensures that the agent thoroughly explores the high-utility outcome space before making significant compromises.

2.2 Frequency-Based Opponent Modeling

To generate counter-offers that are both favorable to Han and likely to be accepted by the opponent, the agent implements a frequency-based opponent model. Throughout the negotiation, the agent catalogs every offer made by the opponent.

When constructing a counter-offer, the agent evaluates the historical frequency of the opponent's proposals. It identifies the opponent's most frequently proposed outcomes and evaluates them against its own utility function $U(o)$. If a historical opponent offer satisfies both the current aspiration target ($U(o) \geq A(t) - 0.05$) and exceeds the reservation value plus a safety margin ($U(o) \geq RV + 0.02$), the agent will opportunistically propose this outcome. This mechanism effectively achieves mutual trade-offs by identifying intersecting interests.

3 Acceptance Mechanism

The acceptance strategy of the Han agent utilizes dynamic, time-dependent utility thresholds rather than a static baseline. The agent evaluates incoming offers against multiple criteria to decide whether to accept, reject, or issue a counter-offer.

3.1 The AC_next Approach

A primary component of the acceptance logic is the *AC_next* (Accept Next) mechanism. Instead of only accepting offers that meet the current mathematical aspiration $A(t)$, the agent projects its *next* planned counter-offer.

Let $o_{current}$ be the opponent's current offer, and o_{next} be the counter-offer the agent is preparing to send. The agent evaluates:

$$U(o_{current}) \geq U(o_{next}) - \delta \tag{2}$$

where δ is a marginal threshold (default $\delta = 0.01$). If the opponent's current offer is infinitesimally close to or better than what the agent plans to ask for next, the agent accepts immediately. This prevents the loss of a near-optimal deal due to rigid adherence to the mathematical curve.

3.2 Deadline Pressure and Strict Walk-Away

As t approaches 1.0, the agent shifts from utility maximization to agreement securitization. When $t \geq 0.90$ (the deadline phase), the agent's acceptance criteria broaden significantly. It will accept any opponent offer that strictly satisfies:

$$U(o_{current}) \geq RV + \text{margin} \tag{3}$$

However, the agent is programmed to never accept poor outcomes. It features an absolute minimum threshold to walk away. At the final ticks ($t \geq 0.98$), if the current offer yields a utility strictly less than the reservation value ($U(o) < RV$), the agent will definitively terminate the session (`END_NEGOTIATION`) rather than concede to an exploitative deal.

4 Multi-Modal Communication Strategy

A defining characteristic of the Han agent is its integration of natural language generation, mapped directly to the negotiation's temporal progress. The communication strategy is divided into three distinct psychological phases, aligning semantic tone with the mathematical bidding state.

4.1 Phase 1: Anchoring and Rapport Building (0% – 30%)

During the early exploration phase, the agent generates text focused on establishing goodwill and setting a collaborative tone. By using phrases such as *"Thank you for joining this negotiation. I look forward to reaching a mutually beneficial agreement,"* the agent attempts to build rapport while simultaneously anchoring the negotiation with its initial, high-utility proposals.

4.2 Phase 2: Logrolling and Tit-for-Tat (30% – 80%)

As the negotiation enters the middle phase and the Boulware curve dictates minor concessions, the communication shifts to highlight these compromises. The agent uses logic-based appeals to encourage reciprocal concessions from the opponent.

The system tracks the utility difference between consecutive proposals. If a concession is made ($\Delta U > 0.05$), the agent outputs: *"I've reconsidered and am moving toward a more balanced position — I hope you'll meet me halfway."* Conversely, if the mathematical model dictates holding firm, the agent switches to a hardline stance: *"I need to maintain my position here — this is close to my walk-away point."* This dynamic toggling simulates a human-like tit-for-tat psychological pressure.

4.3 Phase 3: Urgency and Closing ($>$ 80%)

In the final phase, the communication model leverages the psychology of scarcity and time pressure. As the deadline approaches ($t \geq 0.90$), the agent explicitly signals the impending termination of the negotiation: *"Time is short; this may be our last chance to reach an agreement."*

Furthermore, the agent employs utility transparency to validate its final offers, appending exact utility metrics to its messages (e.g., *"This offer gives me 85% of my maximum utility"*). This transparency acts as a final persuasive push to secure the deal before the strict walk-away threshold is triggered at $t = 0.98$.

5 Algorithm

6 Evaluation

7 Conclusion